DOMAIN EXPECTATIONS

AI/ML

Led by **Kartik Sirohi**

Explore data, understand models, and turn an idea into an experiment.

Python and mathematics

- Lists, dictionaries, functions, and problem solving
- NumPy, Pandas, and Scikit-learn
- Linear algebra and basic calculus

Probability and statistics

- Expectation values, distributions, and hypothesis testing
- Correlation versus causation

Working with data

- Missing values, outliers, data types, and categorical encoding
- Charts, plots, and choosing the right visualization
- Exploratory data analysis, interpreting data, and feature engineering

Models and evaluation

- Linear and logistic regression, decision trees, random forests, and ensemble learning
- SVM and K-means clustering
- Perceptrons, forward and backpropagation, activation functions, and gradient descent
- Bias–variance trade-off and train, test, and validation splits

Submitting an application does not confirm selection. Shortlisting and an offline round follow, and their date, time, and venue will be announced.